LAB 1

Requirements Engineering C UML Use-Case Modelling

PROBLEM STATEMENT #04

CAMPUS AND ACADEMIC OPERATIONS

Student Project Portfolio And Showcase App

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- ## Complete Requirements Table

Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable student teams to create showcase profiles containing project title, team details, abstract summary, demo video link, and GitHub repository URL.	High	Pass: A student team can enter all mandatory project details and save the profile successfully. Fail: The system allows creation when mandatory fields are missing.	Provides a standardized format for presenting student projects.
FR-002	Functional	The system shall validate the submitted GitHub repository URL before publishing a student project profile to the showcase catalog.	High	Pass: A valid GitHub repository URL is accepted and the profile proceeds toward publication. Fail: An invalid or malformed repository URL is accepted.	Ensures showcased projects contain accessible and verifiable repositories.
FR-003	Functional	The system shall enable panel judges to evaluate submitted projects using a predefined digital evaluation rubric and submit their scores.	High	Pass: A judge can enter scores for all required rubric criteria and submit the evaluation. Fail: The system accepts an incomplete evaluation.	Replaces manual evaluation and promotes consistent assessment.
FR-004	Functional	The system shall enable verified public users to cast votes for eligible showcased projects while enforcing anti-sybil voting controls.	High	Pass: A verified user can cast one valid vote according to the configured voting policy. Fail: The system accepts duplicate or unauthorized votes from the same voting identity.	Prevents manipulation of public voting and improves fairness.
FR-005	Functional	The system shall calculate and display the project leaderboard using judge evaluation scores and valid public votes.	High	Pass: The leaderboard reflects newly submitted valid scores and votes. Fail: Invalid or duplicate votes affect the leaderboard.	Provides transparent and updated project rankings.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance C Security	The system shall process public voting requests and leaderboard updates with an API response latency under 300 ms under simulated peak showcase load while enforcing authentication and anti-sybil controls.	High	Pass: Benchmarking tests confirm the target latency and security standards under simulated peak load. Fail: The target latency or required security controls are not met.	Ensures responsive real-time voting and leaderboard updates while protecting voting integrity.

NFR-002	Availability C Reliability	The system shall maintain at least 99.5% availability during the scheduled project showcase period and shall prevent loss of successfully submitted project profiles, evaluations, and valid votes in the event of recoverable system failures.	High	Pass: Monitoring confirms at least 99.5% availability and recovery testing confirms successfully committed data is preserved. Fail: Availability falls below the target or committed data is lost.	Ensures dependable operation during the time-sensitive showcase period.

• UML Use-Case Diagram

The UML use-case diagram for the Student Project Portfolio C Showcase App is shown below.

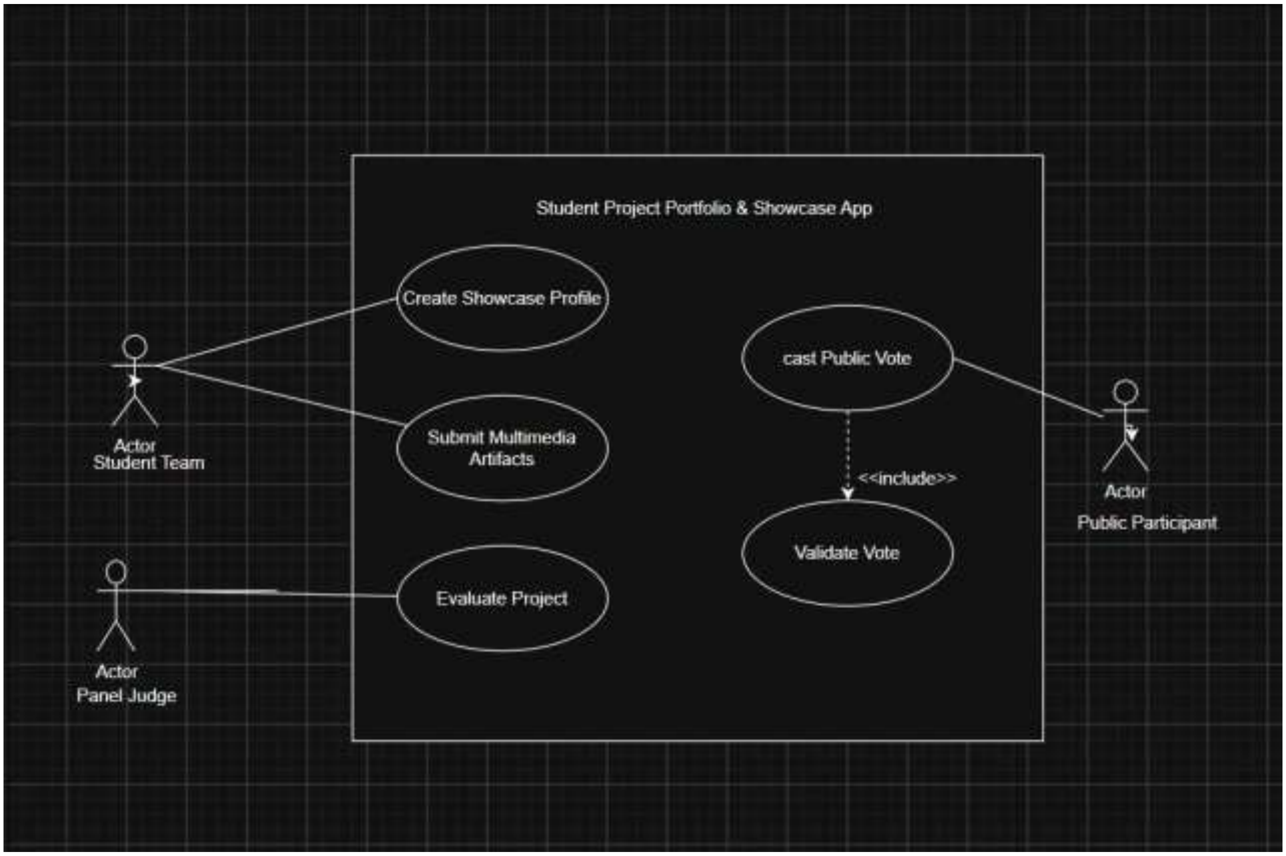

Figure 1: UML Use-Case Diagram — Student Project Portfolio & Showcase App

- ## Use-Case Flow Specification

Field	Specification
Use Case ID	UC-02
Use Case Name	Sumbit Multimedia Artifacts
Primary Actor	Student Team

.1 Preconditions

- The Student Team has access to the showcase application.
- The Student Team is authenticated in the system.
- The showcase submission period is open.
- The Student Team has an existing showcase profile or project entry to which artifacts can be submitted.
- The Student Team has the required multimedia artifacts and valid links/files available.

.2 Postconditions

- The submitted multimedia artifacts are stored and associated with the Student Team's project.
- The system records the artifact submission successfully.
- The submitted artifacts are available for validation and review.
- The Student Team receives confirmation that the multimedia artifacts have been submitted

.3 Main Success Scenario

Step	Main Success Scenario
1	The Student Team logs into the showcase application.
2	The Student Team selects the project for which multimedia artifacts are to be submitted.
3	The system displays the multimedia artifact submission form.
4	The Student Team selects or enters the required multimedia artifacts.
5	The Student Team provides the relevant artifact details or links.
6	The Student Team submits the multimedia artifacts.
7	The system checks whether all mandatory artifact information has been provided.
8	The system validates the submitted artifact formats and links.
9	The system confirms that the submitted artifacts meet the required submission criteria.
10	The system stores the multimedia artifacts and associates them with the project.
11	The system records the successful artifact submission.
12	The system displays a confirmation message to the Student Team.

.4 Alternate Flow A1 — Invalid GitHub Repository URL

- At Step 8, if the system determines that an uploaded artifact has an invalid format, an invalid link, or does not meet the submission criteria, the system displays an error message.
- The system does not record the invalid artifact as a successful submission.
- The Student Team corrects the artifact or replaces the invalid link/file.
- The Student Team resubmits the corrected artifact.
- The system validates the corrected artifact.
- If validation succeeds, the flow continues from **Step 9** of the Main Success Scenario.